Optimal Adaptation under Fiscal Capacity Constraints: Doom and Virtuous Loops in Debt Markets

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Abstract

Physical climate risks such as droughts, floods, and extreme heat are increasingly recognized as systematic risk factors for the real economy and financial markets, yet we know comparatively little about whether and how public policy can reshape their pricing.

Keywords: Physical climate risks, asset pricing, public policy

1. Introduction

Many countries face a policy dilemma: physical climate risk raises expected losses and risk premia in sovereign debt markets, yet the very increase in borrowing costs can compress fiscal space and limit investment in adaptation. This feedback has been described as a "climate–sovereign doom loop." While conceptually compelling, the conditions under which the feedback becomes self-reinforcing—and when it can instead turn into a virtuous cycle—remain insufficiently characterized and empirically operationalized.

This paper develops a tractable macro-finance framework in which the government chooses adaptation investment G_t to manage physical climate risk and financing conditions. Adaptation reduces the persistence and volatility of climate risk and lowers both sovereign and corporate default intensities, compressing public and private spreads. We then introduce an explicit fiscal-capacity (market-access) constraint that endogenizes the shadow price of fiscal tightness, μ_t. This constraint modifies the optimality condition for G_t through a fiscal-space wedge, creating the possibility of non-linear feedback.

Our key theoretical contribution is a set of transparent threshold inequalities that classify the economy into doom-loop versus virtuous-cycle regions. The regime boundary is summarized by $\theta_t \equiv B_t(-\partial s_t^g/\partial G_t)$: when $\theta_t < 1$, a binding constraint raises the effective marginal cost of adaptation, so fiscal tightening reduces G_t^* and reinforces high spreads (doom loop); when $\theta_t > 1$, adaptation improves debt dynamics ($\partial B_{t+1}/\partial G_t < 0$), so fiscal tightness reinforces adaptation incentives (virtuous cycle). A complementary index $\rho_t \equiv \Phi_{s^g}(-\partial s_t^g/\partial G_t) + \Psi_{\bar{s}^c}(-\partial \bar{s}_t^c/\partial G_t)$ captures when adaptation is self-financing on impact. We summarize the regimes in a two-panel map that clarifies policy levers to avoid the doom loop and highlights conditions under which tipping/multiplicity can arise.

Empirically, we bring these predictions to the data using a global panel of sovereign spreads (and CDS where available), public debt, and measures of adaptation/resilience. We estimate how spreads respond to adaptation capacity and interact this response with debt levels and indicators of fiscal tightness to construct empirical counterparts of θ_t and ρ_t. Consistent with the model, we find stronger non-linearities when fiscal capacity is tight, and we provide evidence that the corporate channel strengthens spread compression and expands the virtuous region.

Our results contribute to the sovereign-risk and climate-finance literatures by (i) endogenizing adaptation as an optimal policy choice under fiscal constraints, (ii) offering a regime-classification framework with empirically implementable thresholds, and (iii) highlighting a public–private complementarity channel through which adaptation can generate large fiscal returns.

2. Model Setup

2.1. Economic Environment, Climate Risk, and Adaptation

Time is discrete, $t = 0, 1, 2, \ldots$. Uncertainty is generated by a climate risk factor X_t and other aggregate shocks. Government adaptation investment is denoted by $G_t \geq 0$, which can be interpreted as the flow of public spending on climate adaptation (e.g., large-scale water projects, flood defenses). For now, we treat G_t as exogenous and focus on its asset-pricing implications.

Climate risk dynamics. We model the climate factor X_t as a (possibly persistent) shock that captures the intensity of physical climate risk (e.g., drought risk) at time

t. Its dynamics are given by

$$X_{t+1} = (1 - \phi(G_t))X_t + \sigma(G_t)\,\varepsilon_{t+1}, \tag{1}$$

where ε_{t+1} is an i.i.d. innovation with mean zero and unit variance, $\phi(G_t) \in [0, 1)$ measures the mean-reversion of the climate shock, and $\sigma(G_t) > 0$ is the conditional volatility of X_{t+1}.

We assume:

$$\phi'(G) > 0, \qquad \sigma'(G) < 0. \tag{2}$$

That is, a higher level of adaptation investment G increases the mean-reversion of climate shocks and reduces their conditional variance. Under (1)–(2), G reduces both the persistence and volatility of climate risk.

Aggregate output and tax base. Aggregate output next period is given by

$$Y_{t+1} = \bar{Y} - d_X X_{t+1} + u_{t+1}, \tag{3}$$

where $d_X > 0$ captures the adverse impact of climate risk on output and u_{t+1} is an i.i.d. disturbance. The government collects proportional taxes

$$T_{t+1} = \tau Y_{t+1}, \qquad \tau \in (0, 1).$$

Stochastic discount factor. Asset prices are determined by a stochastic discount factor (SDF) M_{t+1}. We do not commit to a particular preference specification but assume that climate risk is *bad* in the sense that

$$\mathrm{Cov}_t(M_{t+1}, X_{t+1}) > 0. \tag{4}$$

Intuitively, high realizations of X_{t+1} (more severe climate risk) are associated with high marginal utility and thus high M_{t+1}.

2.2. Government Sector and Sovereign Debt Pricing

The government issues one-period defaultable bonds. Let B_t denote the stock of outstanding government debt at time t, and let r_t^f be the risk-free short rate. The government budget constraint is

$$B_{t+1} = (1 + r_t^g)B_t + G_t + I_t - T_t, \tag{5}$$

where r_t^g is the required return on sovereign bonds, G_t is adaptation spending, and I_t collects other government expenditures.

Default intensity. We model sovereign default in reduced form. Let $D_{t+1}^g \in \{0, 1\}$ be the indicator that the government defaults between t and $t + 1$. The (physical) default intensity is

$$\lambda_t^g = \Lambda^g(X_t, B_t, G_t), \tag{6}$$

where $\Lambda^g(\cdot)$ is increasing in climate risk and debt, but decreasing in adaptation:

$$\frac{\partial \Lambda^g}{\partial X} > 0, \quad \frac{\partial \Lambda^g}{\partial B} > 0, \quad \frac{\partial \Lambda^g}{\partial G} < 0. \tag{7}$$

These assumptions capture the idea that severe climate risk and high debt raise sovereign default risk, while greater adaptation investment improves macroeconomic resilience and reduces default intensity.

Sovereign bond pricing and spreads. Consider a one-period sovereign bond with face value one and price P_t^g. Its payoff is $(1 - D_{t+1}^g)$ in units of the numeraire. The no-arbitrage pricing equation is

$$P_t^g = \mathbb{E}_t\Big[M_{t+1}(1 - D_{t+1}^g)\Big]. \tag{8}$$

Define the gross required return

$$1 + r_t^g := \frac{1}{P_t^g}.$$

Let $P_t^f = \mathbb{E}_t[M_{t+1}]$ denote the price of a default-free one-period bond; its return satisfies $1 + r_t^f = 1/P_t^f$. The *sovereign spread* is defined as

$$s_t^g := r_t^g - r_t^f. \tag{9}$$

Under a standard linearization of (8), the sovereign spread can be decomposed into an expected-loss component and a risk-premium component linked to climate risk.[1]

Proposition 1 (Decomposition of sovereign spreads) *Under mild regularity con-*

[1]The exact functional form is not central for our comparative statics. A formal derivation is provided in the Appendix XX.

ditions and a local linearization around the mean, the sovereign spread can be written as

$$s_t^g \approx \underbrace{\lambda_t^{g,\mathbb{P}}}_{\textit{expected loss}} + \underbrace{\beta_X^g(G_t)\,\lambda_X(G_t)}_{\textit{climate risk premium}} + \textit{(other risk premia)}, \tag{10}$$

where $\lambda_t^{g,\mathbb{P}}$ *is the physical default intensity,* $\beta_X^g(G_t)$ *is the sovereign bond's exposure (beta) to the climate factor* X_{t+1}*, and* $\lambda_X(G_t)$ *is the market price of climate risk. Both* $\beta_X^g(G)$ *and* $\lambda_X(G)$ *are functions of the level of adaptation* G *through the climate dynamics (1)–(2) and the SDF covariance (4).*

We next characterize how an increase in adaptation G_t affects the sovereign spread.

Proposition 2 (Climate adaptation and sovereign financing costs) *Suppose that:*

1. Λ^g *satisfies the monotonicity conditions (7);*
2. G *affects the climate dynamics through (1)–(2);*
3. *the SDF satisfies (4).*

Then, holding (X_t, B_t) *fixed, the sovereign spread* s_t^g *is decreasing in* G_t*:*

$$\frac{\partial s_t^g}{\partial G_t} < 0. \tag{11}$$

The total effect can be decomposed into:

$$\frac{\partial s_t^g}{\partial G_t} = \underbrace{\frac{\partial \lambda_t^{g,\mathbb{P}}}{\partial G_t}}_{<0,\ \textit{expected loss effect}} + \underbrace{\frac{\partial \beta_X^g(G_t)}{\partial G_t}\,\lambda_X(G_t) + \beta_X^g(G_t)\,\frac{\partial \lambda_X(G_t)}{\partial G_t}}_{\leq 0,\ \textit{risk premium effect}}.$$

Proof sketch. The expected-loss effect follows directly from (6)–(7). The risk-premium effect arises because G reduces both the variance of X_{t+1} and its persistence, thereby lowering the sovereign bond's exposure $\beta_X^g(G)$, and because the market price of climate risk $\lambda_X(G)$ is decreasing in the volatility of X_{t+1} given (4). A formal derivation is provided in Appendix XX. ■

Corollary 1 (Cross-sectional heterogeneity in sovereign spread relief) *The impact of adaptation investment G_t on sovereign spreads is stronger when:*

1. *the economy is more climate-exposed (higher d_X in (3)), and/or*

2. *the government debt level B_t is higher (so that a given reduction in default intensity generates a larger change in expected loss).*

2.3. Corporate Sector and Credit Spreads

We now introduce a continuum of firms indexed by i. Each firm issues defaultable corporate bonds and is exposed to both climate risk and sovereign risk.

Corporate cash flows and climate exposure. Firm i's cash flow next period is

$$CF_{i,t+1} = \bar{CF}_i - \theta_i X_{t+1} - \eta_i \lambda_t^g + \epsilon_{i,t+1}, \tag{12}$$

where $\theta_i \geq 0$ measures the firm's technological exposure to climate risk (e.g., water-intensive production), $\eta_i \geq 0$ captures its exposure to sovereign risk (e.g., dependence on fiscal transfers or regulated tariffs), and $\epsilon_{i,t+1}$ is an idiosyncratic shock.

Corporate default intensity. Firm i's (physical) default intensity is modeled as

$$\lambda_{i,t}^{c} = \Lambda^{c}(X_t, G_t; Z_i), \tag{13}$$

where Z_i collects firm-level characteristics. We assume

$$\frac{\partial \Lambda^{c}}{\partial X} > 0, \qquad \frac{\partial \Lambda^{c}}{\partial G} < 0, \tag{14}$$

so that worse climate conditions raise default risk, while adaptation investment attenuates it.

Corporate bond pricing and spreads. Consider a one-period zero-coupon bond issued by firm i with price $P_{i,t}^{c}$ and payoff $(1 - D_{i,t+1}^{c})$. The price satisfies

$$P_{i,t}^{c} = \mathbb{E}_t\Big[M_{t+1}(1 - D_{i,t+1}^{c})\Big], \tag{15}$$

where $D_{i,t+1}^{c}$ is the default indicator. The corporate bond return is $1 + r_{i,t}^{c} := 1/P_{i,t}^{c}$, and the corporate credit spread is

$$s_{i,t}^{c} := r_{i,t}^{c} - r_t^{f}. \tag{16}$$

Parallel to Proposition 1 and Proposition 2, we obtain:

Proposition 3 (Decomposition of corporate credit spreads) *Under a local lin-*

earization, the corporate credit spread can be written as

$$s_{i,t}^{c} \approx \underbrace{\lambda_{i,t}^{c,\mathbb{P}}}_{\textit{expected loss}} + \underbrace{\beta_{i,X}^{c}(G_t)\,\lambda_X(G_t)}_{\textit{climate risk premium}} + \textit{(other risk premia)}, \tag{17}$$

where $\lambda_{i,t}^{c,\mathbb{P}}$ is the firm's physical default intensity, $\beta_{i,X}^{c}(G_t)$ is the firm's bond exposure to climate risk, and $\lambda_X(G_t)$ is the same market price of climate risk as in (10).

Proposition 4 (Climate adaptation and corporate financing costs) *If (14) holds and G affects the climate dynamics as in (1)–(2), then, holding (X_t, Z_i) fixed, the corporate credit spread $s_{i,t}^{c}$ is decreasing in adaptation investment:*

$$\frac{\partial s_{i,t}^{c}}{\partial G_t} < 0. \tag{18}$$

The reduction in spreads is stronger for firms with larger climate-exposure coefficients θ_i and sovereign-risk loadings η_i in (12), as summarized in Corollary 2.

Corollary 2 (Cross-sectional predictions for corporate spreads) *For a given change in adaptation investment G_t, the decline in credit spreads $s_{i,t}^{c}$ is predicted to be:*

1. *larger for firms with higher θ_i (more climate-sensitive cash flows, e.g., water-intensive industries); and*

2. *larger for firms with higher η_i (stronger dependence on sovereign risk, e.g., state-owned enterprises or regulated utilities).*

These cross-sectional predictions provide testable implications for the empirical analysis.

2.4. Joint Government–Corporate Financing Conditions

Finally, define the average sovereign spread $\bar{s}_t^g := \mathbb{E}[s_t^g]$ and an aggregate corporate spread $\bar{s}_t^c := \mathbb{E}_i[s_{i,t}^c]$. Combining Propositions 2 and 4, we obtain:

Proposition 5 (Adaptation investment and aggregate financing costs) *Under the above assumptions, an increase in adaptation investment G_t reduces both sovereign and corporate spreads at the aggregate level:*

$$\frac{\partial \bar{s}_t^g}{\partial G_t} < 0, \qquad \frac{\partial \bar{s}_t^c}{\partial G_t} < 0. \tag{19}$$

Moreover, the joint reduction in public and private financing costs can offset part of the contemporaneous fiscal cost of adaptation G_t in the government budget constraint (5).

3. Optimal Adaptation Policy

3.1. Government objective and resource constraint

In the previous subsections, government adaptation investment G_t is taken as exogenous. We now endogenize G_t by specifying a multi-objective government problem in which the policy maker trades off three goals: (i) sustaining aggregate economic activity, (ii) mitigating climate damages, and (iii) improving sovereign and corporate financing conditions.

Let C_t denote aggregate (private) consumption and let $D(X_t)$ capture the flow disutility from climate risk X_t, with $D'(X) > 0$ and $D''(X) \geq 0$.[2] We also let high sovereign and corporate spreads generate additional welfare costs, as they tighten public and private financing constraints. The per-period social welfare is

$$W_t = U(C_t) - \alpha\, D(X_t) - \gamma^g\, s_t^g B_t - \gamma^c\, \bar{s}_t^c K_t, \tag{20}$$

where, $U(\cdot)$ is a standard increasing, concave utility function, $U'(C) > 0$, $U''(C) < 0$; $\alpha > 0$ measures the weight on reducing climate damages; $\gamma^g > 0$ and $\gamma^c > 0$ measure the social costs of high sovereign and corporate financing spreads, respectively; s_t^g is the sovereign spread defined in (9), B_t is government debt; $\bar{s}_t^c$ is an aggregate corporate spread defined in (17); K_t measures the aggregate scale of private borrowing or capital.

The multi-objective nature of the problem is made explicit in (20): the government puts (implicit) weights on consumption, climate damages, and financing conditions. In practice, this is equivalent to a standard single-objective optimization with a weighted composite criterion.

For concreteness, we assume that aggregate consumption is given by

$$C_t = Y_t - G_t - \Phi(B_t, s_t^g) - \Psi(K_t, \bar{s}_t^c), \tag{21}$$

where Y_t is aggregate output defined in (3), G_t is adaptation spending, and $\Phi(\cdot)$

[2] Modeling climate damages partly as utility/amenity damages is standard in the climate-economy literature; see, e.g., Barrage (2020).

and $\Psi(\cdot)$ capture government and private resources absorbed by debt service and financing frictions, with

$$\frac{\partial \Phi}{\partial s^g} > 0, \qquad \frac{\partial \Psi}{\partial \bar{s}^c} > 0.$$

Thus, higher sovereign and corporate spreads tighten the resource constraint by increasing required payments and reducing available consumption.

Government debt evolves according to the budget constraint (5), repeated here for convenience:

$$B_{t+1} = (1 + r_t^g)B_t + G_t + I_t - T_t, \tag{22}$$

with $r_t^g = r_t^f + s_t^g$.

3.2. FOC and characterization of G_t^*

The government chooses a sequence $\{G_t\}_{t\geq 0}$ to maximize the expected discounted sum of social welfare:

$$\max_{\{G_t\}_{t\geq 0}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t W_t, \qquad \beta \in (0, 1), \tag{23}$$

subject to the climate dynamics (1), the output equation (3), the government budget constraint (22), the bond pricing relations (8) and (15), and the definitions of sovereign and corporate spreads in (9) and (17).

Let the state vector be $S_t = (X_t, B_t, \Xi_t)$, where Ξ_t collects other relevant aggregate states (e.g., K_t and exogenous macro shocks). The government's value function satisfies

$$V(S_t) = \max_{G_t \geq 0} \Big\{ W_t(S_t, G_t) + \beta\, \mathbb{E}_t \big[V(S_{t+1}) \big] \Big\}, \tag{24}$$

where S_{t+1} evolves according to the laws of motion implied by the constraints and the climate dynamics (1).

Consider an interior solution and differentiability. The first-order condition (FOC) for G_t can be written as

$$\frac{\partial W_t}{\partial G_t} + \beta\, \mathbb{E}_t \Big[V_X(S_{t+1}) \frac{\partial X_{t+1}}{\partial G_t} + V_B(S_{t+1}) \frac{\partial B_{t+1}}{\partial G_t} + V_\Xi(S_{t+1}) \frac{\partial \Xi_{t+1}}{\partial G_t} \Big] = 0. \tag{25}$$

At time t, (X_t, B_t, K_t, Ξ_t) are predetermined; contemporaneous effects of G_t operate through resource allocation and through prices (spreads) embedded in $\Phi(\cdot)$ and $\Psi(\cdot)$. Using (20)–(21), the marginal effect of G_t on current welfare can be expressed as

$$\frac{\partial W_t}{\partial G_t} = U_C(C_t) \frac{\partial C_t}{\partial G_t} - \gamma^g B_t \frac{\partial s_t^g}{\partial G_t} - \gamma^c K_t \frac{\partial \bar{s}_t^c}{\partial G_t}, \tag{26}$$

where

$$\frac{\partial C_t}{\partial G_t} = -1 + \Phi_{s^g}(B_t, s_t^g) \Big(- \frac{\partial s_t^g}{\partial G_t} \Big) + \Psi_{\bar{s}^c}(K_t, \bar{s}_t^c) \Big(- \frac{\partial \bar{s}_t^c}{\partial G_t} \Big). \tag{27}$$

Equation (27) makes clear that the direct resource cost of adaptation (-1) can be partially offset by contemporaneous spread compression that releases public and private resources.

Combining (25)–(27), the optimality condition equates marginal cost and marginal benefits. We summarize this characterization as follows.

Proposition 6 (Characterization of optimal adaptation) *Suppose* (2) *and* (7) *hold and, analogously, adaptation reduces aggregate corporate spreads so that* $\partial \bar{s}_t^c / \partial G_t <$ 0*. Then the government's optimal adaptation policy* G_t^* *satisfies*

$$U_C(C_t) = \underbrace{U_C(C_t)\Big[\Phi_{s^g}(B_t, s_t^g)\Big(-\frac{\partial s_t^g}{\partial G_t}\Big) + \Psi_{\bar{s}^c}(K_t, \bar{s}_t^c)\Big(-\frac{\partial \bar{s}_t^c}{\partial G_t}\Big)\Big]}_{\text{current benefit: resource release via spread compression}} + \underbrace{\gamma^g B_t\Big(-\frac{\partial s_t^g}{\partial G_t}\Big) + \gamma^c K_t\Big(-\frac{\partial \bar{s}_t^c}{\partial G_t}\Big)}_{\text{current benefit: lower financing costs}}$$
$$+ \underbrace{\beta\,\mathbb{E}_t\Big[V_X(S_{t+1})\frac{\partial X_{t+1}}{\partial G_t} + V_B(S_{t+1})\frac{\partial B_{t+1}}{\partial G_t} + V_\Xi(S_{t+1})\frac{\partial \Xi_{t+1}}{\partial G_t}\Big]}_{\text{intertemporal benefit term}}. \qquad (28)$$

In particular, optimal adaptation trades off the direct resource cost against (i) contemporaneous fiscal returns from spread compression and lower financing costs, and (ii) discounted benefits through next-period states, including future damage mitigation via $\partial X_{t+1}/\partial G_t$ *and debt-sustainability effects via* $\partial B_{t+1}/\partial G_t$.

Proposition 6 highlights that even before introducing an explicit fiscal-capacity constraint, adaptation can be partially self-financing through contemporaneous credit-market improvements. This feature will be central in the regime identification below.

3.3. Fiscal Capacity Constraint and Regime Identification

3.3.1. Fiscal capacity constraint

To capture limited market access and fiscal space, we introduce an explicit fiscal-capacity constraint on next-period debt:

$$B_{t+1} \le \overline{B}(\Xi_t), \qquad (29)$$

where $\overline{B}(\Xi_t)$ is an upper bound that depends on sovereign fundamentals Ξ_t (e.g., institutional quality, resilience, external buffers). Let $\mu_t \geq 0$ denote the associated multiplier. When the constraint binds ($\mu_t > 0$), fiscal tightness has a shadow value, and optimal adaptation is distorted relative to the unconstrained benchmark.

3.3.2. Self-financing and debt-improving indices

The constrained problem modifies the optimality condition through a fiscal-space wedge that depends on how adaptation affects next-period debt. A key object is the marginal effect of G_t on debt dynamics:

$$\frac{\partial B_{t+1}}{\partial G_t} = 1 + B_t \frac{\partial s_t^g}{\partial G_t}. \tag{30}$$

Motivated by (27) and (30), we define two indices:

$$\rho_t \equiv \Phi_{s^g}(B_t, s_t^g)\Big(- \frac{\partial s_t^g}{\partial G_t}\Big) + \Psi_{\bar{s}^c}(K_t, \bar{s}_t^c)\Big(- \frac{\partial \bar{s}_t^c}{\partial G_t}\Big), \qquad \theta_t \equiv B_t\Big(- \frac{\partial s_t^g}{\partial G_t}\Big). \tag{31}$$

The first index ρ_t captures whether adaptation is self-financing on impact through public and private resource release, while the second index θ_t captures whether spread compression is strong enough to improve debt dynamics.

Corollary 3 (Self-financing on impact) *If $\rho_t \geq 1$, then the contemporaneous resource effect satisfies $\partial C_t/\partial G_t \geq 0$ in (27): adaptation is weakly self-financing on impact through reduced public and private resources absorbed by spreads.*

Corollary 4 (Debt-improving adaptation) *If $\theta_t > 1$, then $\partial B_{t+1}/\partial G_t < 0$ in* (30)*: increasing adaptation reduces next-period debt by lowering sovereign spreads sufficiently strongly.*

3.3.3. Doom versus virtuous cycles and tipping

When the fiscal-capacity constraint binds ($\mu_t > 0$), the sign of $\partial B_{t+1}/\partial G_t$ determines whether fiscal tightness acts as a deterrent or an additional incentive to invest in adaptation. Intuitively, if $\theta_t < 1$, increasing G_t raises B_{t+1} mechanically and does not generate enough spread relief to offset the debt effect, so the constraint raises the effective marginal cost of G_t. If $\theta_t > 1$, adaptation reduces B_{t+1}; a binding constraint then strengthens the incentive to invest in G_t.

Proposition 7 (Doom vs. virtuous loop under a fiscal-capacity constraint) *Consider the constraint* (29) *with multiplier $\mu_t \geq 0$ and define θ_t as in* (31)*. When the constraint binds ($\mu_t > 0$), the economy exhibits: (i) a doom-loop region if $\theta_t < 1$, in which tighter fiscal capacity reduces G_t^* and reinforces high spreads; (ii) a virtuous-cycle region if $\theta_t > 1$, in which adaptation improves debt dynamics and fiscal tightness strengthens incentives to invest in G_t. Moreover, tipping/multiplicity is more likely in a neighborhood of $\theta_t \approx 1$ when the feedback between fiscal tightness and policy is sufficiently steep (formalized in Corollary 5).*

Corollary 5 (Tipping and local multiplicity) *Let the constrained best response be summarized by $G_t^* = \mathcal{G}(\mu_t; S_t)$ and fiscal tightness by $\mu_t = \mathcal{M}(B_{t+1} - \overline{B}(\Xi_t))$ locally* *around the boundary, with $\mathcal{M}'(\cdot) > 0$. If the local feedback satisfies*

$$\left|\partial\mathcal{G}(\mu_t; S_t)/\partial\mu_t\right| \left|\mathcal{M}'(\cdot)\right| \left|1 + B_t(\partial s_t^g/\partial G_t)\right| > 1, \tag{32}$$

then small changes in fiscal tightness can trigger large changes in optimal adaptation (tipping), and the model can exhibit locally multiple equilibria.

Figure T.1 summarizes the identification logic in a two-panel regime map. Panel A ($\mu_t = 0$) highlights the role of ρ_t for the contemporaneous fiscal return to adaptation. Panel B ($\mu_t > 0$) highlights the doom versus virtuous boundary at $\theta_t = 1$, with a narrow band around $\theta_t \approx 1$ illustrating heightened tipping risk.

3.3.4. Corporate channel and public–private complementarity

The corporate channel expands the virtuous region through two effects captured by (31). First, by reducing aggregate corporate spreads, adaptation releases private resources absorbed by financing frictions, increasing ρ_t through $\Psi_{\bar{s}^c}(-\partial\bar{s}_t^c/\partial G_t)$. Second, improved private financial conditions can feed back into sovereign risk, strengthening the sovereign spread response $-\partial s_t^g/\partial G_t$ and increasing θ_t. In the regime map, these effects shift the economy upward and to the right, making self-financing on impact more likely and reducing the set of states in which fiscal tightness triggers a doom loop.

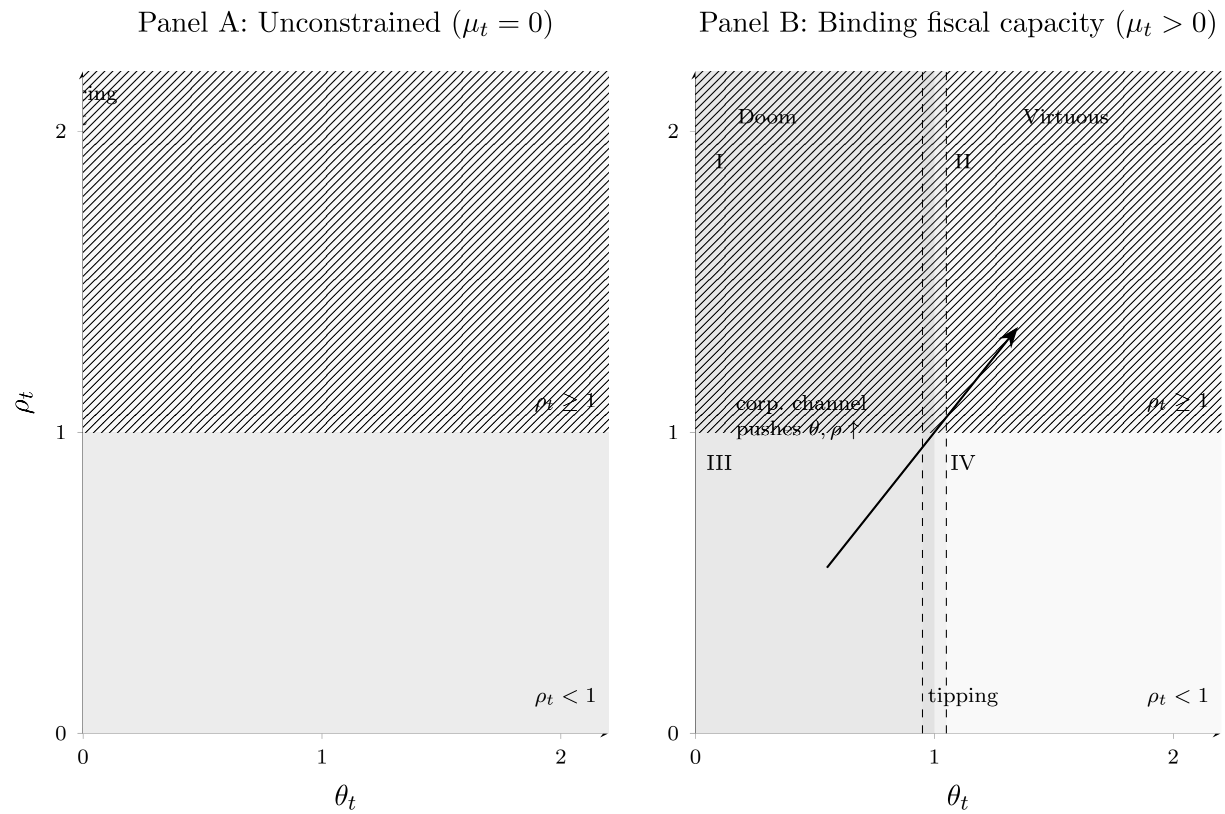

Figure T.1: Regime map accompanying Proposition 7. Panel A ($\mu_t = 0$) emphasizes the self-financing threshold $\rho_t = 1$. Panel B ($\mu_t > 0$) highlights the doom vs. virtuous boundary at $\theta_t = 1$; the shaded band around $\theta_t \approx 1$ is illustrative of where tipping/multiplicity is more likely under strong feedback (Corollary 6). In Panel B, quadrants I–IV follow the standard convention (upper-left, upper-right, lower-left, lower-right).

4. Empirical Evidence

The empirical exercise is intended to validate the sign and sorting implications of the model, rather than to provide a stand-alone causal estimate of fiscal adaptation spending. The estimates are therefore interpreted as theory-guided reduced-form evidence. The main objects are empirical analogues of the model's spread-relief and debt-improving conditions, constructed from country-year variation in sovereign spreads, public debt, and climate-readiness measures.

4.1. Data, Proxies, and Measurement Units

The sample is an unbalanced country-year panel covering 1995–2023. The United States is excluded throughout the empirical analysis. The dependent variable in the spread regressions is the sovereign spread, s_{it}. The adaptation-readiness proxy is G_{it}, constructed from ND-GAIN readiness performance. The vulnerability-performance proxy is X_{it}, constructed from ND-GAIN vulnerability performance. Debt exposure is measured by debt/GDP, B_{it}.

The empirical variables are expressed in analysis units. The adaptation proxy G_{it} is constructed from ND-GAIN readiness performance, while X_{it} is constructed from ND-GAIN vulnerability performance. Higher G_{it} indicates stronger improvements in adaptation readiness. Higher X_{it} indicates lower vulnerability relative to GDP-predicted vulnerability. Sovereign spreads, debt ratios, fiscal ratios, and macroeconomic rates are expressed in share units; for these variables, 0.01 corresponds to one percentage point.[3]

The control vector is denoted by $\mathbf{Z}_{it} = (Z_{1,it}, Z_{2,it}, \ldots, Z_{8,it})'$. The components are real GDP, real GDP growth, CPI inflation, overall balance/GDP, international reserves, government effectiveness, regulatory quality, and terms of trade. The main regressions include country fixed effects and year fixed effects, and report robust t-statistics. Table 1 summarizes the post-preprocessing variables used in the analysis.

[3]For variables expressed in share units, a one-percentage-point change corresponds to a 0.01 change in the regressor. The tables report coefficients in these analysis units.

Table 1: Post-Preprocessing Descriptive Statistics

Block	Variable	N	Mean	SD	Median	Min.	Max.
S	Sovereign spread	1,272	0.024	0.043	0.007	-0.034	0.343
G	Readiness performance	1,798	0.058	0.089	0.048	-0.326	0.303
B	Debt/GDP	1,712	0.586	0.342	0.522	0.039	2.610
X	Vulnerability performance	1,798	-0.040	0.058	-0.047	-0.185	0.295
Z_1	Real GDP	1,791	0.080	0.029	0.077	0.019	0.163
Z_2	Real GDP growth	1,787	0.034	0.036	0.035	-0.145	0.246
Z_3	CPI inflation	1,791	0.057	0.108	0.032	-0.018	1.973
Z_4	Overall balance/GDP	1,697	-0.004	0.034	-0.005	-0.300	0.234
Z_5	International reserves	1,753	0.055	0.128	0.018	0.000	1.527
Z_6	Government effectiveness	1,550	0.006	0.009	0.007	-0.013	0.025
Z_7	Regulatory quality	1,550	0.006	0.008	0.007	-0.013	0.023
Z_8	Terms of trade	1,595	1.006	0.187	0.994	0.319	2.731

Notes: Statistics use the post-preprocessing non-U.S. 1995–2023 panel. Each row is computed on nonmissing observations for the corresponding variable.

4.2. Adaptation and Sovereign Financing Costs

The baseline spread regression tests the sign prediction in Proposition 2:

$$s_{it} = \alpha_i + \tau_t + \beta_G G_{it} + \beta_B B_{it} + \beta_X X_{it} + \mathbf{\Gamma}'\mathbf{Z}_{it} + u_{it}. \tag{33}$$

Table 2 provides reduced-form evidence consistent with Proposition 2. The coefficient on G_{it} is negative and statistically significant, indicating that improvements in the adaptation-readiness proxy are associated with lower sovereign financing costs. The coefficient on B_{it} is positive, and the coefficient on X_{it} is negative. Since higher X_{it} means lower vulnerability relative to GDP-predicted vulnerability, this latter coefficient is consistent with lower spreads among countries with better vulnerability performance.

Table 2: Baseline Fixed-Effects Estimates

Dependent variable	10-year sovereign spread, s_{it}
G_{it}	-0.046***
	(-2.804)
B_{it}	0.043***
	(6.042)
X_{it}	-0.087***
	(-2.626)
Controls $\mathbf{Z}_{it}$	Yes
Country FE	Yes
Year FE	Yes
Sample period	1995–2023
Countries	60
Observations	1,120
Adjusted R^2	0.900

Notes: Robust t-statistics are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

4.3. Heterogeneous Spread Relief and the Empirical Debt-Improving Index

The next step asks whether the spread-relief effect of the readiness-performance proxy varies with debt exposure and vulnerability performance, as suggested by Corollary 1. The full-interaction spread model is

$$s_{it} = \alpha_i + \tau_t + \beta_G G_{it} + \beta_B B_{it} + \beta_X X_{it} + \beta_{GB}(G_{it} \times B_{it}) + \beta_{GX}(G_{it} \times X_{it}) + \mathbf{\Gamma}'\mathbf{Z}_{it} + u_{it}. \quad (34)$$

For the empirical index construction, the expanded full-interaction specification also allows the marginal effect of G_{it} to vary with the control vector:

$$\begin{aligned} s_{it} = {} & \alpha_i + \tau_t + \beta_G G_{it} + \beta_B B_{it} + \beta_X X_{it} \\ & + \beta_{GB}(G_{it} \times B_{it}) + \beta_{GX}(G_{it} \times X_{it}) + \mathbf{\Pi}'(G_{it} \times \mathbf{Z}_{it}) + \mathbf{\Gamma}'\mathbf{Z}_{it} + u_{it}. \end{aligned} \quad (35)$$

Here, $G_{it} \times \mathbf{Z}_{it}$ denotes the vector of interactions between G_{it} and each component of $\mathbf{Z}_{it}$. The implied marginal effect of G_{it} in equation (35) is

$$\frac{\partial s_{it}}{\partial G_{it}} = \beta_G + \beta_{GB} B_{it} + \beta_{GX} X_{it} + \mathbf{\Pi}' \mathbf{Z}_{it}. \tag{36}$$

The implied marginal spread relief and empirical debt-improving index are

$$\widehat{m}_{it}^{G,F} = -\frac{\partial \widehat{s}_{it}}{\partial G_{it}}, \qquad \widehat{\theta}_{it}^{F} = B_{it} \widehat{m}_{it}^{G,F}. \tag{37}$$

The index $\widehat{\theta}_{it}^{F}$ should be interpreted as an empirical proxy for debt-improving capacity, not as a structural estimate of the theoretical θ_t.

Table 3 estimates whether the spread-relief effect of the adaptation proxy varies with debt exposure and vulnerability performance. The negative coefficient on $G_{it} \times B_{it}$ indicates that spread relief is stronger at higher debt levels. Because X_{it} is a vulnerability-performance measure, higher values indicate lower vulnerability relative to GDP-predicted vulnerability. Therefore, stronger marginal relief at lower X_{it} should be interpreted as stronger spread relief among more vulnerable countries. Table 4 then uses the expanded full-interaction specification to summarize the empirical debt-improving index, $\widehat{\theta}_{it}^{F} = B_{it} \widehat{m}_{it}^{G,F}$.

Table 3: Heterogeneous Spread Relief

Dependent variable	10-year sovereign spread, s_{it}			
G_{it}	0.053**	-0.046***	0.058**	0.319***
	(2.116)	(-2.791)	(2.271)	(4.651)
B_{it}	0.062***	0.044***	0.064***	0.070***
	(6.922)	(6.056)	(6.995)	(7.456)
X_{it}	-0.052	-0.084**	-0.043	0.086*
	(-1.562)	(-2.485)	(-1.244)	(1.900)
$G_{it} \times B_{it}$	-0.211***		-0.221***	-0.266***
	(-4.795)		(-4.892)	(-5.723)
$G_{it} \times X_{it}$		0.121	0.268**	0.339**
		(1.196)	(2.302)	(2.255)
Controls $\mathbf{Z}_{it}$	Yes	Yes	Yes	Yes
$G_{it} \times \mathbf{Z}_{it}$ controls	No	No	No	Yes
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Countries	60	60	60	60
Observations	1,120	1,120	1,120	1,120
Adjusted R^2	0.905	0.900	0.906	0.911

Notes: Columns 1–4 report debt heterogeneity, vulnerability-performance heterogeneity, full interaction, and full interaction with $G_{it} \times \mathbf{Z}_{it}$, respectively. Robust t-statistics are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 4: Empirical Debt-Improving Index Descriptive Statistics

Variable	N	Mean	SD	Median	Min.	Max.
$\widehat{\theta}_{it}^{F}$	1,120	0.062	0.151	0.020	-0.094	1.285
$\widehat{m}_{it}^{G,F}$	1,120	0.056	0.106	0.044	-0.184	0.562
$\partial\widehat{s}_{it}/\partial G_{it}$	1,120	-0.056	0.106	-0.044	-0.562	0.184

Notes: Statistics are computed on the expanded full-interaction spread-model estimation sample. The marginal relief variable is $\widehat{m}_{it}^{G,F} = -\partial\widehat{s}_{it}/\partial G_{it}$, and the empirical debt-improving index is $\widehat{\theta}_{it}^{F} = B_{it}\widehat{m}_{it}^{G,F}$.

4.4. Visualizing Spread Relief and the Empirical Index

Figure 1 plots the implied marginal spread relief,

$$-\frac{\partial\widehat{s}_{it}}{\partial G_{it}},$$

from the full-interaction specification. Panel A varies debt/GDP while holding vulnerability performance at its median. Panel B varies vulnerability performance while holding debt/GDP at its median. The confidence intervals are delta-method intervals based on the robust variance-covariance matrix of $(\widehat{\beta}_G, \widehat{\beta}_{GB}, \widehat{\beta}_{GX})$.

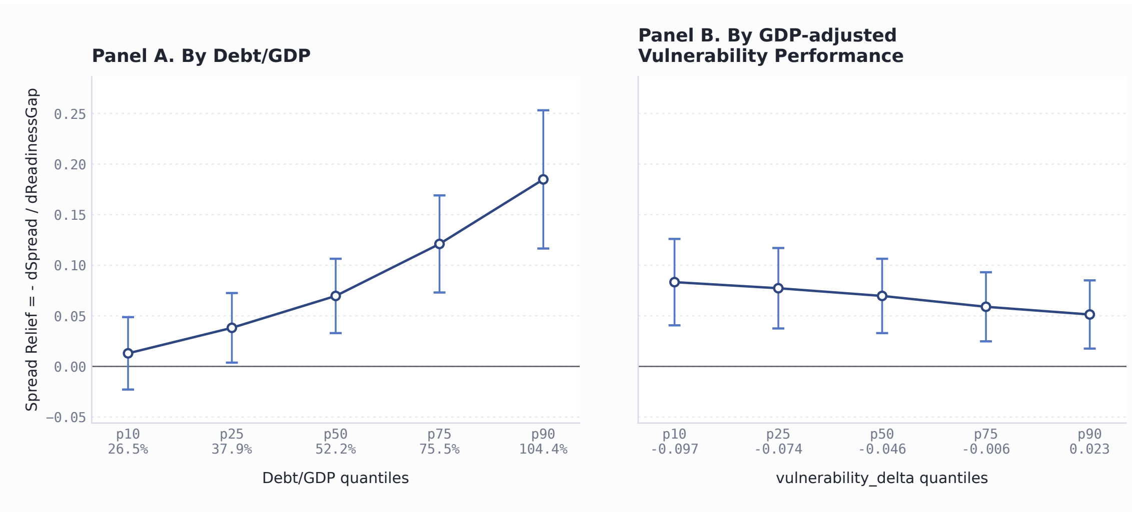

Figure 1: Marginal Spread Relief from the Full-Interaction Spread Model

Notes: Panel A fixes X_{it} at its median. Panel B fixes B_{it} at its median. Higher X_{it} indicates lower vulnerability relative to GDP-predicted vulnerability, so the downward slope in Panel B indicates stronger spread relief among more vulnerable observations.

Figure 2 plots the country-year distribution of $\widehat{\theta}_{it}^F$ and country-level average rankings. The dashed line marks the preferred empirical cutoff selected by the kink specification below. It should not be interpreted as the theoretical threshold $\theta_t = 1$.

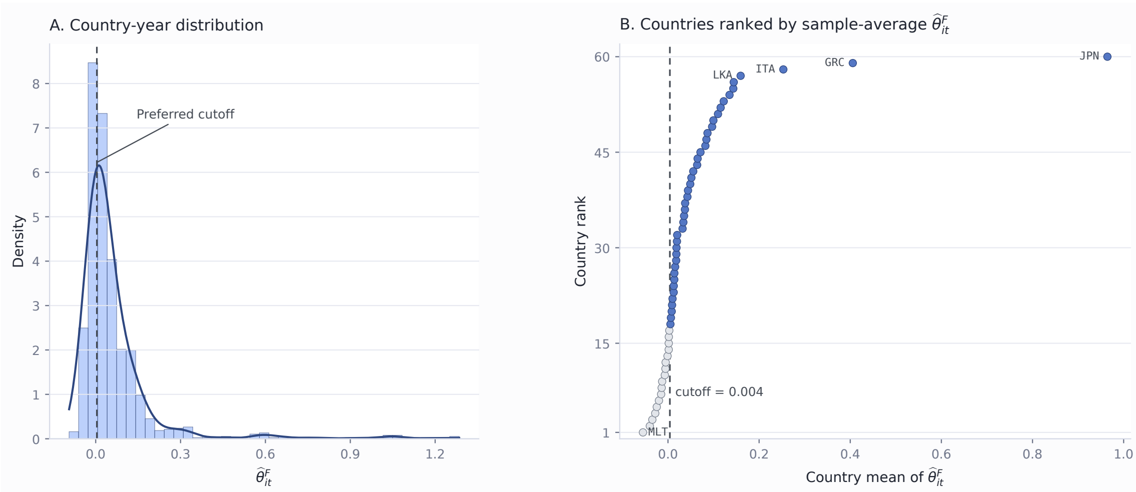

Figure 2: Distribution of the Empirical Debt-Improving Index

Notes: The left panel plots the country-year distribution of $\widehat{\theta}^F_{it}$. The right panel ranks countries by sample-average $\widehat{\theta}^F_{it}$. The dashed line marks the preferred empirical cutoff from the without-theta-controls kink specification.

4.5. Debt Dynamics and the Continuous Theta Interaction

The next set of tests asks whether the readiness-performance proxy is associated with next-period debt changes in a way that varies with the empirical debt-improving index. Define

$$\Delta B_{i,t+1} = B_{i,t+1} - B_{it}. \tag{38}$$

The baseline debt-change regression is

$$\Delta B_{i,t+1} = \alpha_i + \tau_t + \lambda_0 G_{it} + \mathbf{\Gamma}'\mathbf{Z}_{it} + \varepsilon_{it}, \tag{39}$$

and the continuous theta interaction model is

$$\Delta B_{i,t+1} = \alpha_i + \tau_t + \lambda_0 G_{it} + \lambda_1 (G_{it} \times \widehat{\theta}_{it}^F) + \mathbf{\Gamma}' \mathbf{Z}_{it} + \varepsilon_{it}. \tag{40}$$

Table 5 asks whether the marginal effect of the adaptation-readiness proxy on next-period debt changes varies with the empirical debt-improving index. The negative coefficient on $G_{it} \times \widehat{\theta}_{it}^F$ indicates that higher debt-improving capacity is associated with a lower marginal effect of readiness improvements on debt accumulation. This is consistent with Corollary 4, but it should not be read as evidence that high-theta countries mechanically reduce debt in every case.

Table 5: Debt-Change Regressions

Dependent variable	$\Delta B_{i,t+1} = B_{i,t+1} - B_{it}$		
G_{it}	-0.073	-0.042	-0.015
	(-1.463)	(-0.810)	(-0.302)
$G_{it} \times \widehat{\theta}_{it}^F$		-0.886***	-0.310
		(-2.707)	(-0.660)
$\widehat{\theta}_{it}^F$			-0.141*
			(-1.898)
Real GDP	8.143***	7.869***	5.352***
	(5.436)	(5.370)	(2.977)
Real GDP growth	-0.342***	-0.346***	-0.392***
	(-3.715)	(-3.796)	(-4.179)
CPI inflation	-0.173**	-0.174**	-0.158**
	(-2.053)	(-2.060)	(-2.070)
Overall balance/GDP	-0.480***	-0.458***	-0.416***
	(-6.090)	(-5.862)	(-5.204)
International reserves	0.018	0.020	0.019
	(1.299)	(1.456)	(1.366)
Government effectiveness	-1.777*	-1.396	-0.601
	(-1.895)	(-1.470)	(-0.575)
Regulatory quality	1.637*	0.815	-0.405
	(1.857)	(0.907)	(-0.404)
Terms of trade	0.044***	0.037**	0.037***
	(2.845)	(2.484)	(2.584)
Theta main effect	No	No	Yes
Controls $\mathbf{Z}_{it}$	Yes	Yes	Yes
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	1,060	1,060	1,060
Countries	60	60	60
Adjusted R^2	0.425	0.433	0.441

Notes: Columns 1–3 report the baseline model, the continuous theta interaction model, and the full-theta dynamics model, respectively. The full-theta dynamics specification includes $\widehat{\theta}_{it}^F$ as a direct control. Robust t-statistics are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

4.6. Grouped Debt-Change Heterogeneity

Table 6 compares three ways of sorting the sample: by the composite debt-improving index $\widehat{\theta}_{it}^F$, by debt/GDP B_{it}, and by marginal spread relief $\widehat{m}_{it}^{G,F}$. Panel A shows that the coefficient on G_{it} turns from positive in low-$\widehat{\theta}$ groups to negative in high-$\widehat{\theta}$ groups. Panel B shows that sorting by debt/GDP alone does not generate the same clear pattern. Panel C shows a similar ordering when observations are sorted by marginal spread relief, but this sorting omits the debt-exposure component that is central to the theoretical index. Overall, the comparison supports the theoretical composite object $B_{it}\widehat{m}_{it}^{G,F}$, rather than debt exposure alone, as the relevant empirical sorting variable for debt dynamics.

Table 6: Grouped Debt-Change Heterogeneity Regressions

Panel A. Full-theta groups

Dependent variable	$B_{i,t+1} - B_{it}$				
Group	0–20%	20–40%	40–60%	60–80%	80–100%
G_{it}	0.243*	0.057	-0.152	-0.249*	-0.372**
	(1.727)	(0.866)	(-1.550)	(-1.770)	(-2.077)
Controls $\mathbf{Z}_{it}$	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	212	212	212	212	212
Countries	25	33	35	37	22
Adjusted R^2	0.604	0.669	0.499	0.366	0.522

Panel B. Debt-ratio groups

Dependent variable	$B_{i,t+1} - B_{it}$				
Group	0–20%	20–40%	40–60%	60–80%	80–100%
G_{it}	0.060	-0.116	-0.067	0.038	-0.363
	(0.861)	(-1.336)	(-1.021)	(0.291)	(-1.317)
Controls $\mathbf{Z}_{it}$	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	212	212	212	212	212
Countries	28	35	41	32	22
Adjusted R^2	0.466	0.483	0.700	0.394	0.625

Panel C. Marginal-relief groups

Dependent variable	$B_{i,t+1} - B_{it}$				
Group	0–20%	20–40%	40–60%	60–80%	80–100%
G_{it}	0.325**	0.097	-0.041	-0.134	-0.512***
	(2.244)	(0.732)	(-0.426)	(-1.350)	(-3.273)
Controls $\mathbf{Z}_{it}$	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	212	212	212	212	212
Countries	25	36	36	34	24
Adjusted R^2	0.603	0.568	0.518	0.606	0.493

Notes: Each column estimates the next-period change in debt/GDP on current GDP-adjusted readiness performance, G_{it}, within the indicated subgroup. Macro control coefficients and subgroup cutoff values are omitted for compactness. Robust t-statistics are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

4.7. Single-Crossing Kink Test

The single-crossing specification provides the closest empirical analogue to Corollary 4 and its mirror image. It directly tests whether the marginal effect of the adaptation-readiness proxy on next-period debt changes switches sign around an empirical cutoff. This specification is not a standard panel threshold model. It is designed to estimate whether the marginal debt effect of G_{it} crosses zero as a function of $\widehat{\theta}_{it}^F$.

For a candidate cutoff c, the estimated specification is

$$\Delta B_{i,t+1} = \alpha_i + \tau_t + aG_{it}(c - \widehat{\theta}_{it}^F)_+ + bG_{it}(\widehat{\theta}_{it}^F - c)_+ + \mathbf{\Gamma}'\mathbf{Z}_{it} + \varepsilon_{it}. \tag{41}$$

The implied marginal effect is

$$m(\theta; c) = a(c - \theta)_+ + b(\theta - c)_+. \tag{42}$$

The theoretical sign pattern is

$$a > 0, \qquad b < 0.$$

Table 7 selects an empirical cutoff of $c = 0.004123$ in the main without-theta-controls specification. Below the cutoff, the estimated marginal effect is positive, consistent with debt-worsening adaptation. Above the cutoff, the estimated marginal effect is negative, consistent with debt-improving adaptation. Because the empirical objects are proxy-scaled, the estimated cutoff should be interpreted as an empirical index boundary rather than the literal theoretical threshold $\theta_t = 1$. Appendix B reports additional control-set variants and specifications with direct theta controls.

Table 7: Single-Crossing Kink Marginal-Effect Model

Dependent variable	$B_{i,t+1} - B_{it}$
$G_{it}(c - \widehat{\theta}^F_{it})_+$	3.380***
	(2.592)
$G_{it}(\widehat{\theta}^F_{it} - c)_+$	-0.804**
	(-2.464)
Cutoff c	0.004
RSS	1.951
Share below cutoff	0.350
Selection rule	RSS minimum; sign-consistent
Controls $\mathbf{Z}_{it}$	Yes
Country FE	Yes
Year FE	Yes
Observations	1,060
Adjusted R^2	0.435

Notes: The exact RSS-selected cutoff is $c = 0.0041231435658208$. The model includes country and year fixed effects and $\mathbf{Z}_{it}$ controls, but does not include direct theta controls. Robust t-statistics are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$.

Figure 3 plots the continuous marginal-effect curve implied by equation (42). The left side of the cutoff corresponds to the debt-worsening region, while the right side corresponds to the debt-improving region.

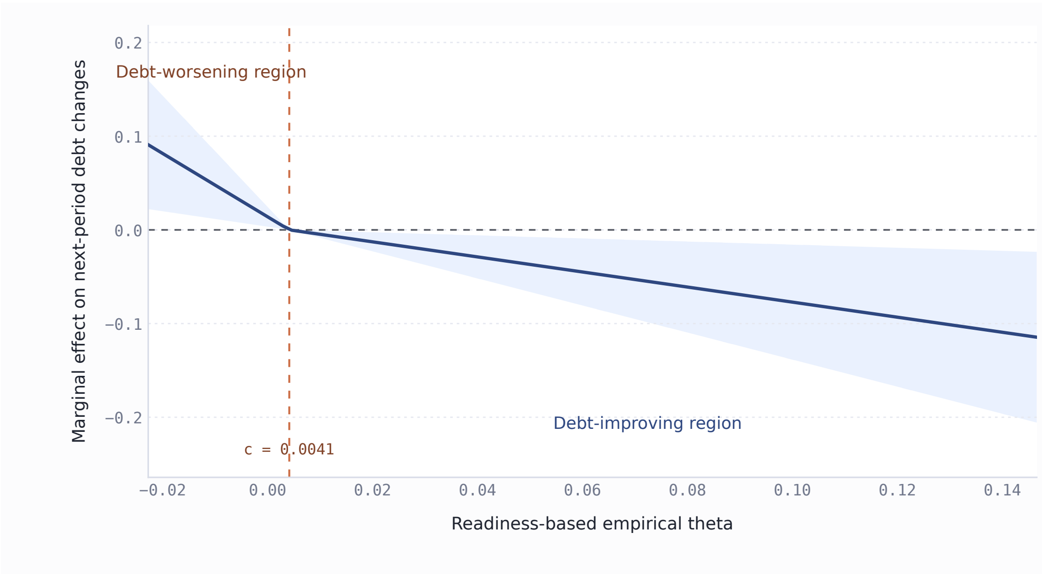

Figure 3: Continuous Kink Marginal Effect without Theta Controls
Notes: The figure plots $m(\theta; c) = a(c - \theta)_+ + b(\theta - c)_+$ from the single-crossing kink model without theta controls. Positive values imply that readiness improvements are associated with higher next-period debt changes; negative values imply lower next-period debt changes. The vertical dashed line marks the empirical cutoff c.

4.8. Country Screens Around the Preferred Cutoff

Finally, Table 8 translates the preferred empirical cutoff into an illustrative country-level screen. This exercise is descriptive rather than classificatory. It shows which countries are persistently located on the high-relief or low-relief side of the empirical index. The country screen should be read as a diagnostic application of the empirical index, not as a structural country typology. Vietnam, Indonesia, and Colombia appear as low-debt, high-relief cases under the screening rule, while Malta appears as a high-debt, low-relief case.

Panel A. Low debt, theta always above the cutoff, and top-quartile marginal relief

Table 8: Country Screens Around the Preferred Cutoff

Country	ISO3	Sample period	Years	Mean debt/GDP	Mean theta	Mean $\widehat{m}^{G,F}$
Vietnam	VNM	2007–2023	17	39.50%	0.065	0.164
Indonesia	IDN	2003–2023	21	33.72%	0.055	0.156
Colombia	COL	2002–2023	21	45.82%	0.064	0.134

Panel B. High debt, theta always below the cutoff, and bottom-quartile marginal relief

Country	ISO3	Sample period	Years	Mean debt/GDP	Mean theta	Mean $\widehat{m}^{G,F}$
Malta	MLT	2010–2023	14	53.91%	-0.055	-0.107

Notes: The preferred cutoff is $c = 0.004123$. Screening thresholds use country-level means: the mean-debt median is 53.06 percent, the bottom quartile of mean $\widehat{m}_{it}^{G,F}$ is 0.003, and the top quartile is 0.112. The always-above and always-below conditions are evaluated year by year.

References

Barrage, L. (2020). Optimal dynamic carbon taxes in a climate–economy model with distortionary fiscal policy. *Review of Economic Studies*, 87(1):1–39.

A. Proofs

A.1. Proof of Proposition 7: Doom vs. virtuous loop under a fiscal-capacity constraint

Consider the government's constrained one-period problem at date t:

$$V^c(S_t) = \max_{G_t \geq 0} \Big\{ W_t(S_t, G_t) + \beta\, \mathbb{E}_t\big[V^c(S_{t+1})\big] + \mu_t\big(\overline{B}(\Xi_t) - B_{t+1}\big) \Big\}, \tag{A.1}$$

where $\mu_t \geq 0$ is the multiplier on the fiscal-capacity constraint (29), together with the complementarity conditions

$$\mu_t \geq 0, \qquad \overline{B}(\Xi_t) - B_{t+1} \geq 0, \qquad \mu_t\big(\overline{B}(\Xi_t) - B_{t+1}\big) = 0. \tag{A.2}$$

For an interior choice of G_t, the first-order condition is

$$\frac{\partial W_t}{\partial G_t} + \beta\, \mathbb{E}_t\left[V^c_X(S_{t+1}) \frac{\partial X_{t+1}}{\partial G_t} + V^c_B(S_{t+1}) \frac{\partial B_{t+1}}{\partial G_t} + V^c_\Xi(S_{t+1}) \frac{\partial \Xi_{t+1}}{\partial G_t} \right] - \mu_t \frac{\partial B_{t+1}}{\partial G_t} = 0. \tag{A.3}$$

Relative to the unconstrained first-order condition (25), the binding fiscal-capacity constraint adds the wedge $-\mu_t(\partial B_{t+1}/\partial G_t)$.

Using the debt law of motion in (30) and the definition of θ_t in (31),

$$\frac{\partial B_{t+1}}{\partial G_t} = 1 + B_t \frac{\partial s^g_t}{\partial G_t} = 1 - \theta_t. \tag{A.4}$$

Hence (A.3) can be written as

$$H(G_t, \mu_t; S_t) \equiv \mathcal{F}(G_t; S_t) - \mu_t\big(1 - \theta_t\big) = 0, \tag{A.5}$$

where

$$\mathcal{F}(G_t; S_t) \equiv \frac{\partial W_t}{\partial G_t} + \beta\,\mathbb{E}_t\left[V_X^c(S_{t+1})\frac{\partial X_{t+1}}{\partial G_t} + V_B^c(S_{t+1})\frac{\partial B_{t+1}}{\partial G_t} + V_\Xi^c(S_{t+1})\frac{\partial \Xi_{t+1}}{\partial G_t}\right], \tag{A.6}$$

and

$$\theta(G_t; S_t) \equiv B_t\Big(- \frac{\partial s_t^g}{\partial G_t}\Big). \tag{A.7}$$

Assume the strengthened local second-order/regularity condition

$$H_G(G_t^*, \mu_t; S_t) < 0, \tag{A.8}$$

Applying the implicit function theorem to (A.5) yields

$$\frac{\partial G_t^*}{\partial \mu_t} = -\frac{H_\mu(G_t^*, \mu_t; S_t)}{H_G(G_t^*, \mu_t; S_t)} = \frac{1 - \theta_t}{H_G(G_t^*, \mu_t; S_t)}, \tag{A.9}$$

where $\theta_t \equiv \theta(G_t^*; S_t)$ is evaluated at the optimum. Because the denominator is negative by (A.8), the sign of $\partial G_t^*/\partial \mu_t$ is the opposite of the sign of $1 - \theta_t$.

This proves Proposition 7. $\square$

A.2. Proof of Corollary 5: Tipping and local multiplicity

The local equilibrium choice of adaptation solves the fixed-point problem

$$G_t = \mathcal{T}(G_t; S_t) \equiv \mathcal{G}\big(\mathcal{M}\big(B_{t+1} - \overline{B}(\Xi_t)\big); S_t\big)\,. \tag{A.10}$$

Differentiating (A.10) with respect to G_t and applying the chain rule gives, at a local fixed point G_t^*,

$$\mathcal{T}_G(G_t^*; S_t) = \frac{\partial \mathcal{G}(\mu_t; S_t)}{\partial \mu_t} \cdot \mathcal{M}'(\cdot) \cdot \frac{\partial B_{t+1}}{\partial G_t}. \tag{A.11}$$

Using (30),

$$\frac{\partial B_{t+1}}{\partial G_t} = 1 + B_t \frac{\partial s_t^g}{\partial G_t}, \tag{A.12}$$

so

$$\big|\mathcal{T}_G(G_t^*; S_t)\big| = \big|\partial \mathcal{G}(\mu_t; S_t)/\partial \mu_t\big| \big|\mathcal{M}'(\cdot)\big| \big|1 + B_t(\partial s_t^g/\partial G_t)\big|. \tag{A.13}$$

Therefore condition (32) is exactly the statement that

$$\big|\mathcal{T}_G(G_t^*; S_t)\big| > 1. \tag{A.14}$$

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This proves Corollary 5. □

B. Additional Empirical Results

This appendix reports control-set variants for the debt-change and single-crossing kink specifications. The main text treats the without-theta-controls kink model with $\mathbf{Z}_{it}$ controls as the preferred specification. The no-theta-controls kink specification displays the clearest single-crossing pattern. Specifications with direct theta controls preserve the positive low-theta effect but attenuate the high-theta debt-improving effect, suggesting that the evidence is strongest when $\widehat{\theta}_{it}^{F}$ is used as the sorting index rather than absorbed directly as an additional control.

Table B.1: Debt-Change Model Control-Set Variants

Panel A. Baseline debt-change model

Dependent variable		$B_{i,t+1} - B_{it}$	
Control set	$\mathbf{Z} + B + X$	$B + X$	None
G_{it}	-0.024	-0.069	-0.030
	(-0.569)	(-1.513)	(-0.577)
B_{it}	-0.093***	-0.098***	
	(-3.839)	(-4.607)	
X_{it}	-0.143	-0.492***	
	(-1.241)	(-4.306)	
Controls $\mathbf{Z}_{it}$	Yes	No	No
Observations	1,060	1,060	1,060
Adjusted R^2	0.449	0.372	0.327

Panel B. Continuous theta interaction model

Dependent variable		$B_{i,t+1} - B_{it}$	
Control set	$\mathbf{Z} + B + X$	$B + X$	None
G_{it}	-0.016	-0.056	0.010
	(-0.370)	(-1.166)	(0.182)
$G_{it} \times \widehat{\theta}^F_{it}$	-0.250	-0.432	-1.236***
	(-0.576)	(-1.009)	(-3.504)
B_{it}	-0.086***	-0.087***	
	(-2.675)	(-3.183)	
X_{it}	-0.118	-0.457***	
	(-0.974)	(-3.981)	
Controls $\mathbf{Z}_{it}$	Yes	No	No
Observations	1,060	1,060	1,060
Adjusted R^2	0.449	0.373	0.346

Panel C. Full-theta dynamics model

Dependent variable		$B_{i,t+1} - B_{it}$	
Control set	$\mathbf{Z} + B + X$	$B + X$	None
G_{it}	-0.021	-0.077	0.038
	(-0.473)	(-1.615)	(0.728)
$G_{it} \times \widehat{\theta}^F_{it}$	-0.133	0.062	-0.302
	(-0.267)	(0.138)	(-0.674)
$\widehat{\theta}^F_{it}$	-0.044	-0.188***	-0.194***
	(-0.584)	(-2.727)	(-3.320)
B_{it}	-0.076**	-0.035	
	(-2.369)	(-1.130)	
X_{it}	-0.146	-0.538***	
	(-1.088)	(-4.311)	
Controls $\mathbf{Z}_{it}$	Yes	No	No
Observations	1,060	1,060	1,060
Adjusted R^2	0.449	0.383	0.367

Notes: Each specification includes country and year fixed effects. Robust t-statistics are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$.

For the $\mathbf{Z}_{it} + B_{it} + X_{it}$ version reported in Table B.2, the without-theta-controls kink specification is

$$\Delta B_{i,t+1} = \alpha_i + \tau_t + aG_{it}(c - \widehat{\theta}_{it}^F)_+ + bG_{it}(\widehat{\theta}_{it}^F - c)_+ \\ + \delta_B B_{it} + \delta_X X_{it} + \mathbf{\Gamma}'\mathbf{Z}_{it} + \varepsilon_{it}. \tag{B.1}$$

Table B.2: Single-Crossing Kink without Theta Controls: Control-Set Variants

Dependent variable		$B_{i,t+1} - B_{it}$	
Control set	$\mathbf{Z} + B + X$	$B + X$	None
$G_{it}(c - \widehat{\theta}_{it}^F)_+$	3.402**	3.583	4.343**
$p(a > 0)$	0.010	0.052	0.005
$G_{it}(\widehat{\theta}_{it}^F - c)_+$	-0.154	-0.345	-1.061***
$p(b < 0)$	0.360	0.205	0.001
RSS cutoff	0.004	-0.003	0.006
Sign-consistent	Yes	Yes	Yes
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	1,060	1,060	1,060
Adjusted R^2	0.452	0.375	0.351

Notes: These variants keep the main without-theta-controls kink structure but vary the additional controls. The preferred main-text cutoff remains the $\mathbf{Z}_{it}$-controls specification. Reported p-values are one-sided tests for the sign restriction. *** $p < 0.01$, ** $p < 0.05$.

For the $\mathbf{Z}_{it} + B_{it} + X_{it}$ version corresponding to Table B.3, the kink specification with direct theta controls is

$$\Delta B_{i,t+1} = \alpha_i + \tau_t + aG_{it}(c - \widehat{\theta}_{it}^F)_+ + bG_{it}(\widehat{\theta}_{it}^F - c)_+ \\ + \rho_1\widehat{\theta}_{it}^F + \rho_2(\widehat{\theta}_{it}^F - c)_+ + \delta_B B_{it} + \delta_X X_{it} + \mathbf{\Gamma}'\mathbf{Z}_{it} + \varepsilon_{it}. \tag{B.2}$$

Table B.3: Single-Crossing Kink with Theta Controls

Dependent variable	$B_{i,t+1} - B_{it}$
$G_{it}(c - \widehat{\theta}^F_{it})_+$	3.560***
	(3.163)
$G_{it}(\widehat{\theta}^F_{it} - c)_+$	-0.113
	(-0.233)
Direct theta controls	Yes
Cutoff c	0.004
RSS	1.889
Share below cutoff	0.350
Selection rule	RSS minimum; sign-consistent
Controls $\mathbf{Z}_{it}$	Yes
Country FE	Yes
Year FE	Yes
Observations	1,060
Adjusted R^2	0.452

Notes: This appendix version includes $\widehat{\theta}^F_{it}$ and $(\widehat{\theta}^F_{it} - c)_+$ as direct controls. The low-theta coefficient remains positive and significant, but the high-theta debt-improving coefficient is much weaker. Robust t-statistics are reported in parentheses. *** $p < 0.01$.

For the $\mathbf{Z}_{it} + B_{it} + X_{it}$ column reported in Table B.4, the estimated with-theta-controls variant uses the same control-augmented kink equation:

$$\begin{aligned} \Delta B_{i,t+1} = {} & \alpha_i + \tau_t + aG_{it}(c - \widehat{\theta}^F_{it})_+ + bG_{it}(\widehat{\theta}^F_{it} - c)_+ \\ & + \rho_1 \widehat{\theta}^F_{it} + \rho_2(\widehat{\theta}^F_{it} - c)_+ + \delta_B B_{it} + \delta_X X_{it} + \mathbf{\Gamma}'\mathbf{Z}_{it} + \varepsilon_{it}. \end{aligned} \tag{B.3}$$

Table B.4: Single-Crossing Kink with Theta Controls: Control-Set Variants

Dependent variable		$B_{i,t+1} - B_{it}$	
Control set	$\mathbf{Z} + B + X$	$B + X$	None
$G_{it}(c - \widehat{\theta}^F_{it})_+$	3.303*	2.576	5.873***
$p(a > 0)$	0.044	0.075	0.000
$G_{it}(\widehat{\theta}^F_{it} - c)_+$	-0.101	0.018	-0.020
$p(b < 0)$	0.416	0.516	0.482
RSS cutoff	-0.012	-0.004	-0.004
Sign-consistent	Yes	No	Yes
Direct theta controls	Yes	Yes	Yes
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	1,060	1,060	1,060
Adjusted R^2	0.456	0.401	0.387

Notes: These variants keep direct theta controls and vary the additional controls across $\mathbf{Z}_{it} + B_{it} + X_{it}$, $B_{it} + X_{it}$, and no controls. Reported p-values are one-sided tests for the sign restriction. *** $p < 0.01$, * $p < 0.10$.